

Internship Report – Day 9

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 9 Overview:

Today, I worked extensively on the ANPR project by exploring the use of YOLOv8 for number plate detection and understanding the role of data augmentation in improving model training. I attempted to use Roboflow to prepare and export the dataset in YOLOv8 format but found that downloading trained models requires a paid plan. To overcome this, I switched to Google Colab for training and installed the required libraries like ultralytics and torch, though I faced several compatibility and import errors. After troubleshooting, I managed to unzip the dataset and began setting up the training environment, but encountered a missing image path error due to incorrect dataset directory structure. Overall, I made good progress in setting up the training pipeline despite technical hurdles.

Tools Used Today:

colab, GEMINI, SONNET, Roboflow, Chatgpt, hugging face, youtube.

Task Given:

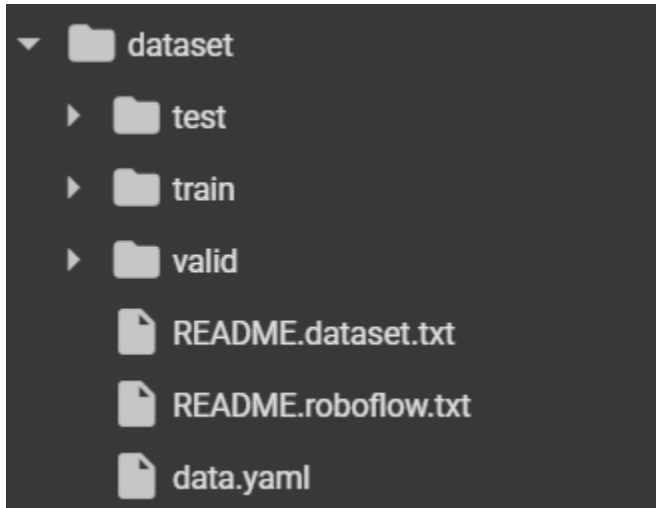
TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

Approaches:-

For the ANPR project, I adopted a structured workflow involving dataset preparation, model training setup, and environment troubleshooting. I first selected and augmented the dataset using Roboflow, then exported it in YOLOv8 format for training. I planned to train the model in Google Colab using the Ultralytics YOLOv8

library. After unzipping the dataset and installing dependencies, I faced version issues with PyTorch and errors related to model imports. I resolved these by updating installations and restarting the runtime. Finally, I attempted to train the model, but ran into a missing file path error, which I identified as a directory structure mismatch. This iterative process helped solidify my understanding of model training and dataset management.

Attachments:-



```
!pip install ultralytics
from IPython.display import clear_output
clear_output()
print("✅ Ultralytics YOLOv8 installed!")
```

✅ Ultralytics YOLOv8 installed!

```
# STEP 1: Uninstall conflicting versions (if any)
!pip uninstall -y torch torchvision torchaudio

# STEP 2: Install compatible PyTorch version with CUDA support
!pip install torch==2.0.1 torchvision==0.15.2 torchaudio==2.0.2 --index-url https://download.pytorch.org/whl/cu118

# STEP 3: Install Ultralytics (YOLOv8)
!pip install ultralytics

# STEP 4: Test import
try:
    from ultralytics import YOLO
    print("✅ YOLO is ready to use!")
except Exception as e:
    print("❌ Error importing YOLO:", e)
```

```

Found existing installation: torch 2.6.0+cu124
Uninstalling torch-2.6.0+cu124:
  Successfully uninstalled torch-2.6.0+cu124
Found existing installation: torchvision 0.21.0+cu124
Uninstalling torchvision-0.21.0+cu124:
  Successfully uninstalled torchvision-0.21.0+cu124
Found existing installation: torchaudio 2.6.0+cu124
Uninstalling torchaudio-2.6.0+cu124:
  Successfully uninstalled torchaudio-2.6.0+cu124
Looking in indexes: https://download.pytorch.org/whl/cu118
Collecting torch==2.0.1
  Downloading https://download.pytorch.org/whl/cu118/torch-2.0.1%2Bcu118-cp311-cp311-linux_x86_64.whl (2267.3 MB)
    2.3/2.3 GB 426.8 kB/s eta 0:00:00
Collecting torchvision==0.15.2
  Downloading https://download.pytorch.org/whl/cu118/torchvision-0.15.2%2Bcu118-cp311-cp311-linux_x86_64.whl (6.1 MB)
    6.1/6.1 MB 92.2 MB/s eta 0:00:00
Collecting torchaudio==2.0.2
  Downloading https://download.pytorch.org/whl/cu118/torchaudio-2.0.2%2Bcu118-cp311-cp311-linux_x86_64.whl (4.4 MB)
    4.4/4.4 MB 89.3 MB/s eta 0:00:00
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch==2.0.1) (3.18.0)
Requirement already satisfied: typing-extensions in /usr/local/lib/python3.11/dist-packages (from torch==2.0.1) (4.13.2)
Requirement already satisfied: sympy in /usr/local/lib/python3.11/dist-packages (from torch==2.0.1) (1.13.1)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch==2.0.1) (3.4.2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.11/dist-packages (from torch==2.0.1) (3.1.6)
Requirement already satisfied: triton==2.0.0 (from torch==2.0.1)
Collecting triton==2.0.0 (from torch==2.0.1)
  Downloading https://download.pytorch.org/whl/triton-2.0.0-1-cp311-cp311-manylinux2014_x86_64_manylinux_2_17_x86_64.whl (63.3 MB)
    63.3/63.3 MB 11.1 MB/s eta 0:00:00
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torchvision==0.15.2) (2.0.2)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from torchvision==0.15.2) (2.32.3)
Requirement already satisfied: pillow<8.3.0, >=5.3.0 in /usr/local/lib/python3.11/dist-packages (from torchvision==0.15.2) (11.2.1)
Requirement already satisfied: make in /usr/local/lib/python3.11/dist-packages (from triton==2.0.0->torch==2.0.1) (3.1.6)
Collecting lit (from triton==2.0.0->torch==2.0.1)
  Downloading https://download.pytorch.org/whl/lit-15.0.7.tar.gz (132 kB)
    132.3/132.3 kB 10.8 MB/s eta 0:00:00
Preparing metadata (setup.py) ... done

```

```

from google.colab import files
uploaded = files.upload()

import zipfile
import os

# Unzip the dataset
for file in uploaded.keys():
    if file.endswith(".zip"):
        with zipfile.ZipFile(file, 'r') as zip_ref:
            zip_ref.extractall("dataset")
print("✅ Dataset unzipped!")

```

```

from ultralytics import YOLO

# Path to your dataset config (data.yaml file)
data_yaml_path = "dataset/data.yaml" # Replace with your actual path

# Initialize YOLO model (you can use yolov8n.pt for faster, smaller model or yolov8s.pt for a medium-sized model)
model = YOLO("yolov8n.pt") # Change to yolov8m.pt or yolov8s.pt if needed
from ultralytics import YOLO

model = YOLO("yolov8n.pt") # or yolov8s.pt if you want better accuracy

model.train(data="/content/dataset/data.yaml", epochs=30, imgsz=640)

# After training, the model will automatically save the best model
print("✅ Training complete! Check the saved model: best.pt")

```

Ultralytics 8.3.133 Python-3.11.12 torch-2.0.1+cu118 CPU (Intel Xeon 2.20GHz)

Extraction complete

7720 files extracted to Google Drive. 2 files were skipped

[View Files in Drive](#)

